

14. Life below water: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
15. Life on land: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
16. Peace, justice, and strong institutions: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17. Partnerships for the goals: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Provide a discussion on how your project relates to or addresses any of the SDGs.

Intellectual Property Rights

An undergraduate student on a teaching course will normally have limited opportunity to create IP relating to their studies. However, on Computer Science courses the student may be involved in project work. Depending on the circumstances, the project work may involve working with other students, academic staff or other employees of the University. In these conditions IP created may be jointly owned between the collaborating parties. Many of the research groups are funded by research grants or industrial sponsorship with an expectation that arising IP will be protected and where possible commercially exploited. Consequently, the student may be requested to sign an agreement relating to confidentiality and IP prior to being allowed to work with the research group. The IP & Commercialisation team will set up the short confidentiality and IP agreement based on a standard template and can answer any queries regarding the agreement which a student might have. The confidentiality provisions will not affect the ability of the student to submit a written project report and in the event that the student has contributed to IP which is later commercially exploited they will receive the same revenue share as they would if they were an employee of the University.

For more details, please refer to the "[Provision and Processing of Intellectual Property Rights for Students and Graduates at The University of Nottingham](#)" document. In order to discuss your specific situation, please contact the IP management group via tto@nottingham.ac.uk by sending an email with the subject line of "Computer Science: Final Year L3 project IP query".

Important Dates and Deliverables

The following list provides provisional important dates for the project students, including approximate deadlines for the submission of deliverables. Please refer to the Moodle COMP3003 project webpage for the exact dates when the academic year starts.

- **Early September:** Moodle Project Allocation System (PAS) opens for student project sign-up before the academic year starts.
- **Early October:** Deadline for students to agree on a project topic with a supervisor via Moodle Project Allocation System (PAS). Students who have not found a supervisor by this date will be assigned to available supervisors by the project coordinator.
- **Mid-October:** Deadline for submitting your detailed project proposal including a detailed project plan (up to four A4 sides) to your supervisor. Your supervisor will give you formative feedback on your project proposal and plan; you must revise it according to your supervisor's feedback.

- **Late October/Early November:** Deadline for submitting your
 - final project proposal (in PDF) via Moodle for supervisor approval, and
 - preliminary [Research Ethics](#) checklist after discussing the ethical issues with your supervisor in relation to your project via the Moodle project webpage. Should the preliminary checklist indicate that full ethics clearance is required; the full clearance process can be completed later if appropriate. However, note that **no** work requiring ethics clearance must be undertaken prior to having obtained clearance.
- **Around Christmas:** Deadline for submitting your interim report (in PDF). The interim report comprises **10%** of the mark for the module.
- **Around Easter (Following Year):** Deadline for electronic submission of your dissertation. **Dissertations will be marked based on electronic submissions.** Details of the suggested dissertation structure can be found in the Dissertation section and dissertation marking criteria in the Assessment section. The dissertation comprises **75%** of the mark for the module.
- **Just Before the Assessments Period (Following Year):** Final Presentation/Demonstration Days. You will give a final presentation/demonstration of your project. Assessment is based on the quality of work (e.g., software) and the presentation and communication. The demonstration/presentation mark comprises **15%** of the mark for the module.
- In the Assessments Period (Following Year): Release of demo marks.
- After the Assessments Period (Following Year): Release of provisional dissertation marks.

Project Proposal

The project proposals will not be assessed, however, they will be taken into account in the assessment of the interim reports. The project proposals should not exceed four A4 sides (2000 words) excluding the cover page, and containing the following main components:

- **Cover/title page:** This should include the project title, your name, ID, email, and name of the programme of your study (e.g., CS with AI).
- **Background and motivation:** Briefly describe the background to the project, importance/need of the area, and motivation for carrying out the proposed work.
- **Aim(s) and Objectives:** The aim is usually a single sentence describing at a high level what the point of the project is and what will be achieved. The objectives are sub-components of the general aim, detailing the individual aspects which need to be achieved in order to deliver the aim(s).
- **Work plan:** Describing the tasks to be carried out. The time plan should be realistic and should take into account other commitments such as exam periods, holidays, etc. The inclusion of a Gantt chart is strongly recommended as a visual representing the project schedule.
- **Bibliography** containing some key publications that are either explicitly referred to in the text.

Examples of Good Proposals

- [Vehicle Hot Spot Identification on Data Streams, Rebecca Tickle \[PDF\]](#)
- [The Swift Instant Messenger, Marcin Wisniewski \[PDF\]](#)

- [The Impact of Social Settings and Virtual Humans on Facial Expressions, In Response to Taste Stimuli, Humaira_Ahmed \[PDF\]](#)
- [Monadic streams library, Chris Purdy \[PDF\]](#)

Interim Report

An interim report consists of an outline with a progress update and draft content. The following is a typical interim report structure provided as guidance and this varies depending on the nature of the project. A brief description of each component can be found in the Dissertation section below. Please discuss your interim report with your supervisor before submission.

- **Introduction**
- **Motivation**
- **Related work**
- **Description of the work**
- **Methodology**
- **Design**
- **Implementation (if there is any)**
- **Progress**
 - **Project management** covering the tasks as a part of your **work plan as presented in your final proposal** and progress as well as how time and resources are managed. The need for reviewing the tasks and explanation of/reasons for any adjustment made to the future work plan. Inclusion of a Gantt chart is strongly recommended as a visual indicating the progress of the project.
 - **Contributions and reflections** providing the details of your achievements and contributions up to date as well as a personal reflection on the plan and your experience of the project (a critical appraisal of how the project has been progressing). This section should also explain how you have considered the “bigger picture” within which your work is situated in terms of **LSEPI** (refer to the Laws, Social, Ethical and Professional Issues in Projects section above), highlighting which issues are the most significant for your project, and how you will address them in the remainder of your project.
- **Bibliography**

There is a **suggested** maximum limit of 5000 words without exceeding 15 pages (A4 sides) for the interim report, including the bibliography and excludes cover/front pages (e.g., abstract, acknowledgment, table of contents).

Examples of Good Interim Reports

The following samples are the main bodies of "good" interim reports selected from various years:

- [Predicting Solar Irradiance, Timothy Cargan \[PDF\]](#)
- [Scheduling and Optimisation of IT Support Incidents, Aidan Reed \[PDF\]](#)
- [Machine Learning for Market Prediction, Jay Mehta \[PDF\]](#)
- [A Data-Driven Approach to Bus Timetable Optimisation Recommendations, Jonathan Foot \[PDF\]](#)
- [Novel Approaches for Optimal Control of Battery Energy Storage Systems, Eskil Sulen Gjerde \[PDF\]](#)

- [Improving Selection Hyper-Heuristics Based on Analyses of their Search Behaviour using HyVis, Jack Cakebread \[PDF\]](#)
- [Predicting Key Presses over Video Calls using an Audio Side-Channel Attack and Machine Learning, Adam Masters \[PDF\]](#)

Dissertation

A demo video needs to be submitted as a part of the dissertation, which will also get assessed.

The following is a typical dissertation structure. However, this is for guidance only: depending on the type and specific aspects of your project, you may find it appropriate to structure your dissertation in a more or less different way. Discuss with your supervisor.

- **Title page** with a signed declaration that the dissertation is your own work [[WORD template](#)]. If submitting electronically, your initials would suffice as a signature. You are reminded of the University's guidelines on [Academic Integrity and Misconduct](#).
- **Abstract** giving a short overview of the work in your project.
- **Table of contents** giving page numbers for all major section headings.
- **Introduction** setting out the aims and objectives of your project, **explaining the overall intention of the project and specific steps that will be taken to achieve that intention.**
- **Motivation** explaining the problem being solved and its importance or need for it.
- **Related work** explaining what your project does that is new or is better than existing work in the same field
- **Description of the work** explaining what your project is meant to achieve, how it is meant to function, e.g., perhaps even a functional specification for a software-oriented project
- **Methodology** describing the (theoretical/experimental/analytical/numerical/software development-based/research-based, etc.) methodologies/techniques/tools/algorithms/technologies/etc. that are relevant to the project topic justifying the choices made for the project, where possible with supporting evidence derived from the existing work.
- **Design** containing a comprehensive description of the design chosen, how it addresses the problem, and why it is designed the way it is
- **Implementation** containing a comprehensive description of the implementation of your software, including the language(s) and platform chosen, problems encountered, any changes made to the design as a result of the implementation, etc.
- **Evaluation** explaining how your software/approach was tested (using different datasets or in different environments), statistical evaluation of performance, results of user evaluation questionnaires, etc.
- **Summary and Reflections** including a discussion of results in a wider context (considering other work).
 - **Project management covering** the tasks as a part of your work plan and progress as well as how time and resources are managed.
 - **Contributions and reflections** providing the details of your achievements and contributions including innovation, creativity and novelty (if there is any) as well as a personal reflection on the plan and your experience of the project (a critical appraisal of how the project went). This section should also explain how you have considered the “bigger picture” within which your work is situated in terms of **LSEPI** (refer to the Laws, Social, Ethical and Professional Issues in Projects

section above). You should present your reflections on each issue, including why they are (or are not) significant for your project.

- **Bibliography** containing a *complete* list of books and other publications that are either explicitly referred to in the text, or which are recommended for further reading on the topic
- **Appendices**, e.g., User Manuals, supporting evidence for claims made in the main part of the dissertation (e.g. a copy of a user evaluation questionnaire), samples of test data, etc. Note that Appendices are optional

There is a maximum limit of 15,000 words without exceeding 40 pages (A4 sides) for the **main body** of the dissertation that will be submitted in PDF. This limitation includes the bibliography and excludes cover/front pages (e.g., abstract, acknowledgment, table of contents) and excludes the appendices, listing of any codes or any other supporting documentation.

Note: Your dissertation **should not exceed the word and page limits**. You do not have to use up your word limit to get a good grade; never 'pad out' your dissertation - which is likely to get penalised. The length of the report is one of the marking criteria that will be considered by the markers and you can always violate a requirement at your own risk, as you could lose marks.

Dissertation – Demo Video

You should prepare a **1 to 3-minute** long presentation and record it using a (screen capture) software tool as a demo video. The maximum limit for the duration of the demo video should not be exceeded. You can always violate a requirement at your own risk as you could lose marks. A demonstration video should be encoded and submitted as 1280x720 H264 video in an MP4 wrapper. This is a standard format that can be generated from most video software on all platforms. Videos submitted in other formats will be rejected. The focus of your demo video should be the outcome of your project. For example, for a software-oriented work, you should discuss the resultant software product illustrating its (main) features, how it works and your contribution in the demo video. For a research-oriented project, the novelty of the work, your contribution, and results should be discussed. As a general guide for the demo videos,

- prepare the narrative for the demo video before the recording
- you can edit pieces of narrative to produce the final demo video
- remember having the title of your project on the first video frame and **stating** the title of your project and your name (do not use or mention your ID).
- speak clearly
- use audio-visual aids as appropriate illustrating crucial features of your work.

The following sample demo videos are provided just as a guide.

- [Vehicle Hot Spot Identification on Big Data Streams \[MP4\]](#)
- [Multimodal Function Optimization using Differential Evolution \[MP4\]](#)
- [Creating a Unity Tool to Make Finite Spaces Infinitely Explorable with Overlapping Geometry \[MP4\]](#)
- [Novel Approaches for Optimal Control of Battery Energy Storage Systems \[MP4\]](#)

Examples of Good Dissertations

The following samples are the main bodies of "good" dissertations selected from various years:

- [A Study of Parallelisation of Pattern Search and Covariance Local Search, Yuyang Zhou \[PDF\]](#)
- [A feet prediction and correction system for a 3D character in Unity, Zain Ali \[PDF\]](#)
- [Computer-aided Expert System for Breast Cancer Diagnosis, Xin Yu Liew \[PDF\]](#)
- [Novel Approaches for Optimal Control of Battery Energy Storage Systems, Eskil Sulen Gjerde \[PDF\]](#)
- [Programming with Monsters, Christopher Purdy \[PDF\]](#)
- [Semantic segmentation and quantification of spontaneous intracerebral haemorrhage on CT images using deep learning, Yong En Kok \[PDF\]](#)
- [Fugue: A Friendly Functional Programming Language with Holes, Zachariah Garby \[PDF\]](#)

Dissertation Submission Instructions

Your dissertation needs to be submitted electronically via Moodle. You **do not need** to submit "large" library files for the code part of your submission; **only the source code** of your implementation (along with instructions regarding which libraries should be included in the environment for the execution of your code in a README file) would be sufficient if the overall size of the code exceeds the limit. Please follow the precise submission instructions on Moodle.

Your dissertation will be marked based on your electronic submission. Late submission (of any one component: the main body of the dissertation, demo video, supplementary material, and code) is subject to the standard University penalty of **5%** reduction per working day. For more information, please refer to the [Policy on Late Submission of Assessed Materials](#).

Electronic submission

You need to make your electronic submission in **four** parts. Please submit all files listed below using Moodle (under Assignments choose the link to Dissertation), **by the deadline**

(i) main body of your dissertation as a PDF (the only accepted format!). When you include a cover or front page, please provide only your ID and not your name/surname.

(ii) demo video: recording of the demo as an MP4 (the only accepted format!)

(iii) other supplementary material (such as, appendices, data, lengthy proofs, length empirical results or any other supporting documentation) as one file (.zip or .tar .gz)

(iv) your code (preferably the source code and a readme file providing links to the libraries for creating the executable that will be used during the demo, including any end-user documentation) as a compressed file (.zip, .jar or .tar.gz).

Please contact the Project Coordinator if you have problems accessing Moodle or have a good reason to request an extension. Your supervisor cannot give you an extension (neither can the module convenor without an approved extenuating circumstances form).

Demonstration

An essential component of the project is to present your work and discuss it in a professional manner. The duration of a demonstration/presentation is fixed and includes a question and answer session. The exact duration for demonstrations is announced via the Moodle project web page. Depending on the nature of your project (e.g, research or software), an appropriate combination of presentation/demonstration would be required. Please take advice from your supervisor on this. The demonstration/presentation/viva is usually organised by the Project Co-ordinator and announced via the Moodle project web page. Normally, your supervisor and second marker who is a panel member attend the demonstration. Where relevant the external sponsor or anyone else the supervisor thinks is appropriate could attend.

Warning: If you do not attend your demonstration/presentation, you could fail the whole project.

The demonstration/presentation gives the markers of your project the chance to gather further information on the quality of your work (e.g., research, software product), as well as to observe how well you can demonstrate/present it. Considering the limited demonstration time, you should prepare carefully and do a mock-up demonstration/presentation before the date, perhaps with your supervisor or friends. You may need to be selective on what you are going to demonstrate/present, however, make sure that you cover the key objectives and outcomes of your project as well as your contributions and originality (if there is any) of your work. You need to make appropriate prior arrangements for the demonstration/presentation. As general guidance for the demonstration/presentation,

- remember introducing yourself
- be enthusiastic about your own work – you worked hard for a year to achieve the project goals
- speak clearly making eye contact with the *audience*
- use visual aids as appropriate
- be ready to explain the limitations of your work, difficulties that you encountered, and what further work could be done.

Assessment

Projects will be assessed based on the following aspects of your project:

- Knowledge and understanding
- Critical thinking skills
- Technical achievement
- Technical writing skills
- Professionalism

The marking criteria that will be used **as a guideline** in the assessment of each project component is explained below.

Interim Report Marking Criteria

The interim report is marked by the supervisor only **[10%]**. Considering that the work is in progress, a subset of the relevant dissertation marking criteria (e.g., technical writing skills, appropriateness of the aim and objectives, knowledge on the subject area and topic, critical

thinking skills, progress and plan) will be used in the assessment. Evidence should be available of substantial work having taken place, leading to good progress towards the end of the semester against the project plan the students were to have handed in by the deadline. Any reason for being behind plan should be explained through good reflective reasoning – not just I did not work hard enough.

More details on the marking criteria can be found in the **interim report marking rubric** [\[PDF\]](#). This rubric provides a set of guidelines outlining the typical requirements of specific grade band for the interim reports. It is not necessary that a project exhibit all the criteria listed in each grade given below to fall within that band. The characteristics of each band are merely representative of that grade. The balance between various aspects of the particular work will be considered during assessment. The supervisor will provide a written feedback justifying the final mark.

Demonstration Marking Criteria

The demonstration is marked by the supervisor **[15%]** and project panel member based on two main criteria:

- **[10%]**: quality of work (e.g., quality of software, degree of originality, and individual contribution), and
- **[5%]**: quality of presentation and communication (e.g., time management, quality of content, degree of clarity, quality of structure and style, ability to make the demo interesting, ability to respond to questions and discuss).

The supervisor will provide agreed written feedback justifying the given mark.

Dissertation Marking Criteria

Your supervisor will grade your dissertation based on the following criteria and this will be moderated by a panel of markers **[75%]**.

- Knowledge and understanding
 - Breadth of knowledge on the subject area and knowledge of the chosen specific topic as the focus
 - Understanding of the (software development/research) methodologies / techniques / tools related to the project (e.g., specification, design, synthesis, etc.)
 - Quality & usage of sources
- Critical thinking skills
 - Extent of the justification of the method(s) (theoretical / experimental / analytical / numerical / design / objectives). E.g. related to theory / literature / standard practices / performance requirements / specification
 - Ability to describe relevant results
 - Discussion and interpretation of result/ evaluation of software (own project): ability to sum up and interpret them, showing an awareness of limitations of own work.
 - Discussion and interpretation of results/ evaluation of software in wider context: ability to interpret own results in wider context.
 - Reflection on original plan

- Technical achievement
 - Quality of software (work/product/ implementation/ research)
 - Quality of supporting documentation (e.g., testing results, input/output data, user manual, comments etc.)
 - Attainment of agreed appropriate objectives and requirements
- Technical writing skills
 - Section headings
 - Visual elements (graphs, figures, tables): range of types of visual elements, quality and adequate presentation (labels, captions)
 - Degree of consistency of the formatting of the document (font size, references list, use of bold/italic, layout, spacing, etc.)
 - Structure of sentences and division of text into paragraphs
 - Use of in-text citations to support development of argument
 - Accuracy of the grammar
 - Typos
 - Style (use of technical vocabulary)
 - Length of report
- Professionalism
 - Ability to self-manage
 - Professional approach to project and time-management
 - Innovation and specific individual contributions
 - Demonstration video

More details on the marking criteria can be found in the project **dissertation marking rubric** [\[PDF\]](#). This rubric provides a set of guidelines outlining the typical requirements of a specific grade band for the dissertations. It is not necessary that a project exhibit all the criteria listed in each grade given below to fall within that band. The characteristics of each band are merely representative of that grade. The balance between various aspects of the particular work will be considered during assessment. The supervisor will provide written feedback justifying the final mark.

Resits

Resit projects get assessed based on 80% dissertation and 20% demonstration. The supervisor acts as the first marker for assessing the dissertation and demonstration components, while a second marker is also assigned for marking moderation. The same deadline for the dissertation submission will be followed in the same academic year when the resit takes place. Depending on the circumstances, there is flexibility to arrange "remote"/online demonstrations over an appropriate medium, such as MS Teams.